Where Does High α Emerge From: A Nonlinear Portfolio Construction Method Based on Adaptive Portfolio Trees and Adaptive Pruning Algorithm

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Abstract

This study proposes and validates a framework that integrates Adaptive Portfolio Trees (APT), which capture complex factor interactionsand an Adaptive Pruning (AdaP) algorithm. The model first systematically generates a large number of candidate portfolios via the APT structure, then employs the AdaP algorithm to select a sparse set of final strategies. Based on a 60-month out-of-sample test, the APT+AdaP strategy delivers outstanding performance, achieving an annualized Sharpe ratio of 2.75 and a cumulative return exceeding 1500%. The strategys excess returns primarily stem from the systematic capture of a core path characterized by "low B/M + large size + deep value". This structure is not a simple superposition of traditional factors, but a stable and economically meaningful deep factor interaction pattern automatically identified by the adaptive portfolio tree in the high-dimensional factor space. This study provides a new perspective for systematically discovering interpretable sources of alpha in the Chinese market.

Keywords: Asset Pricing; Asset Portfolio Trees; Adaptive Pruning; Machine Learning; Long-Term Trading Strategies

I Introduction

1. Background and Motivation

The core task of modern asset pricing lies in identifying risk factors that can systematically explain cross-sectional variations in stock returns. Since the Fama-French three-factor model established the foundation, academia has uncovered hundreds of potential pricing factors, forming what is termed the "factor menagerie" (Cochrane and John, 2011). While expanding market understanding, this phenomenon poses significant challenges: within high-dimensional factor spaces, how can one effectively distinguish genuine pricing influences from data mining artefacts, and construct robust portfolios? Traditional linear multi-factor models struggle to capture complex non-linear interactions between factors, particularly the phenomenon of "conditional alpha". Consequently, developing a new paradigm capable of systematically exploring nonlinear relationships and performing robust model selection in high-dimensional spaces has become a frontier issue in asset pricing research.

This paper aims to propose a research framework combining systematic exploration with intelligent screening. Using Chinese A-share market data, we validate its effectiveness in constructing investable, interpretable, and high-performing quantitative strategies. Through rigorous out-of-sample validation, we provide empirical support for the framework's practical utility, offering a transparent and viable pathway for asset pricing research within high-dimensional, non-linear environments.

2. Literature review

In recent years, machine learning techniques have emerged as powerful new tools for addressing the aforementioned challenges. A series of papers in leading journals has demonstrated their significant potential for predicting stock returns and identifying new factors. For instance, a landmark study compared the performance of various machine learning approaches, including deep learning, gradient-boosted trees, and penalized regression, in predicting stock returns. It was found that these methods significantly outperformed traditional linear models, with non-linear models (such as neural networks and tree models) demonstrating particularly pronounced advantages. The value of machine learning was emphasized to lie not only in prediction, but also in its capacity to handle complex interactions among numerous factors (Gu et al., 2020).

Against this backdrop, tree models have received a great deal of attention thanks to their nonlinear structure and interpretability. The novel concept of an 'asset pricing forest' has been introduced, using random forests to identify the most important pricing factors and their interactions. This research found that the importance of many traditional factors depends on the values of other variables (Bryzgalova et al., 2025). However, ensemble models such as random forests often operate as a 'black box',

making the final composite weights difficult to understand and implement intuitively.

To strike a balance between model complexity and interpretability, a key theme has emerged: using sparsity to identify the most critical drivers. For example, Lasso regression has been used to filter out macro- and micro-level characteristics that significantly impact asset pricing, revealing that only a handful of features exhibit consistent out-of-sample predictive power (Freyberger et al., 2020). This focus on a parsimonious factor structure is echoed in other research, which argues that cross-sectional return variations can be explained by a small number of risk factors. By using Principal Component Analysis (PCA) and sparsity techniques, these critical underlying factors can be effectively identified from a multitude of securities (Serhiy Kozak and Santosh, 2020).

In a parallel line of research focusing on dynamic relationships, 'instrumented principal component analysis' (IPCA) was proposed to construct time-varying risk premiums by incorporating firm characteristics. This method extends traditional dimensionality reduction techniques and further highlights the complex, dynamic interplay between factors and firm-specific attributes (Kelly et al., 2019). Together, these cutting-edge studies point towards a trend: future asset pricing research requires a unified framework that can systematically characterize nonlinear interactions and perform robust, sparse model selection within high-dimensional spaces.

3. **Marginal contribution**

Recent structural empirical work highlights a critical limitation in standard investment-based asset pricing models: while they explain cross-sectional average returns well, they fail to capture the time-series dynamics of stock returns, suggesting that rigid functional forms may miss key nonlinearities or omitted state variables(Belo et al., 2024). Motivated by these findings, this paper departs from restrictive structural assumptions and proposes a data-driven, two-stage framework integrating the adaptive portfolio tree and aptive pruning algorithm to systematically explores the high-dimensional, nonlinear interaction space of firm characteristics to construct sparse and robust portfolios. Comprehensive analysis is employed using data from China's A-share market spanning from 1995 to 2024. The primary contributions of this study are as follows:

Firstly, systematically constructing a non-linear strategy space. Based on the portfolio tree concept, and considering long-term data integrity, economic significance, local computing power, and feature consistency (i.e., all being fundamental features), three types of factorsmarket capitalization, book-to-market ratio, and return on equityare selected for in-depth sorting. A total of 1,555 candidate strategies (including all possible factor splitting sequences and intermediate nodes) are constructed to system-

atically capture high-order factor interactions.

Secondly, achieving robust strategy screening and portfolio optimization. Faced with such a large-scale strategy space, this paper uses an AdaP framework based on LARS/Lasso, combined with time-series segmentation, cross-validation, and grid search, to achieve automated and robust sparse strategy selection.

Thirdly, providing rigorous out-of-sample empirical evidence: The final strategy is compared with two benchmarks (equal-weighted full market, i.e., the root node; and equal-weighted selected nodes) to demonstrate its significant advantages.

II Data acquisition and processing

1. **Data acquisition**

   The empirical analysis in this paper is based on monthly individual stock-level data from the Chinese stock market. The raw data originates from the CSMAR database, providing unprocessed transaction data, risk-free interest rate data, and financial statement data. The final dataset utilized in this study was processed from this source, covering the time period from 1995 to 2024 and presenting as stock-month time panel data. The data fields obtained are described in **Table 1**, which details the variables utilized in this study. Panel A presents the original monthly transaction and financial data fields extracted from the CSMAR database for this research. Panel B elaborates on the core derived variables constructed from the raw data, which form the foundation for building the adaptive portfolio tree and the adaptive pruning model.

2. **Data processing**

   Upon obtaining raw transaction data, risk-free interest rate data, and financial statement data, this paper systematically preprocessed the data to construct stock-by-month panel data suitable for asset pricing research. First, monthly individual stock transaction data across the entire market was aggregated with the corresponding year's risk-free interest rate data to calculate monthly excess returns for each stock. Subsequently, financial data was time-aligned with transaction data at the stock level to ensure the use of the most recently disclosed financial statement data, thereby avoiding forward-looking bias. Subsequently, based on the merged panel data, firm characteristics such as size (log of total market capitalization), book-to-market ratio, and return on equity were calculated. Stocks with missing values exceeding 20% were either excluded or imputed, whilst robust treatments were applied to address obvious non-zero or missing cases, thereby reducing the interference of outliers on results. Concurrently, a rolling window method was employed to compute return volatility. Besides, the continuous variables underwent 1% and 99% tail trimming. Through this data preprocessing procedure, a cleaned monthly dataset covering the entire market from 1995 to 2024 was ultimately obtained, suitable for both cross-sectional and time-series asset pricing tests.

Table 1: Variable Definitions and Descriptions

Panel A: Raw Data Fields from CSMAR	
Variable	Source/Details
stkcd	Unique A-share stock code for a listed company.
trdmnt	Year and month of the observation.
mclsprc	Closing price on the last trading day of the month.
mretwd	Monthly stock return, considering reinvestment of cash dividends.
msmvttl	Total shares outstanding multiplied by the closing price at month-end.
msmvosd	Tradable A-shares outstanding multiplied by the closing price at month-end.
mnshrtrd	Cumulative number of shares traded during the month.
mnvaltrd	Cumulative total value of shares traded during the month.
Risk-free Rate	Monthly risk-free rate, typically derived from the one-year government bond yield.
Total Assets	Total assets as reported in the balance sheet for the fiscal period.
Shareholder's Equity	Total equity attributable to shareholders of the parent company, from the balance sheet.
Net Profit	Net profit attributable to shareholders of the parent company, from the income statement.

Panel B: Key Constructed Variables		
Variable	Description	Formula
Excess Return	The stock's monthly investment return above the risk-free rate.	Monthly Return with Dividend − Risk-free Rate
Size	A logarithmic measure of the company's market capitalization.	ln(Monthly Total Market Value)
B/M	A valuation ratio comparing a company's book value to its market value.	$\frac{\text{Shareholder's Equity}_{t-1}}{\text{Monthly Total Market Value}_t}$
ROE	An accounting measure of a firm's profitability.	$\frac{\text{Net Profit}_{t-1}}{\text{Shareholder's Equity}_{t-1}}$
Return Volatility	A measure of the uncertainty of a stock's returns.	Calculated as the standard deviation of monthly returns over the past N months (e.g., N=12 or 24).

3. **Descriptive statistics**

Table 2: Descriptive Statistics

Variables	N	Mean	Std	P25	P50	P75	Max	Min
Excess Return	751647	0.008	0.132	-0.072	-0.003	0.074	0.468	-0.304
Size	751647	15.428	1.091	14.692	15.281	16.024	18.831	13.191
ROE	751647	0.036	0.088	0.009	0.032	0.070	0.292	-0.463
BM	751647	0.004	0.004	0.002	0.003	0.004	0.024	0.000
Volatility	57609	0.124	0.055	0.086	0.113	0.150	0.334	0.041

Note: The continuous variables underwent 1% and 99% tail trimming. This table reports the summary statistics of the data we used. Variable definitions are provided in Table 1.

III Models

This section details the methodology employed to construct the adaptive portfolio trees (APT) model utilized in this research. We first introduce the two-stage core architecture of the APT model: the systematic generation of portfolio trees and the adaptive pruning

algorithm. Finally, we present the model's performance during the out-of-sample testing period and conduct an in-depth analysis of the characteristics of the final portfolio.

1. **Model construction: Adaptive Portfolio Tree(APT)**

   The core model of this study is the adaptive portfolio tree (APT), a two-stage framework for constructing portfolios. It is designed to robustly screen effective yet sparse investment strategies from a high-dimensional factor interaction space.To systematically explore these interactions, we first construct an extensive library of candidate portfolios. Specifically, for a given set of factors (size, value and profitability), we generate all possible depth-4 ordered sequences. For each annual dataset and factor sequence, we build a portfolio tree through sequential ordering.

   (a) Root node: contains all stocks for that year;

   (b) First-level division: Based on the first factor in the sequence (e.g. 'value'), the stocks in the root node are divided into two sub-nodes (high-value and low-value portfolios) according to their median value;

   (c) Subsequent divisions: For each sub-node generated at the previous level, the median division within each group is repeated based on the next factor in the sequence.

   This process is repeated until a predetermined tree depth is reached (depth 4 in this study). Each node in the tree represents a unique portfolio that has been screened using multiple factors. We remove duplicate portfolios and nodes (and their child nodes) with an abnormal number of stocks (failing to meet subsequent splitting requirements), and the generate time series of portfolio returns and node descriptions. By traversing all years and all factor sequences, a high-dimensional matrix is generated, which is the input for the subsequent algorithmic phase. A total of 1,555 candidate portfolio nodes are finally formed.

2. **Adaptive-Pruning Algorithm(AdaP)**

   The AdaP algorithm proposed in this paper is a portfolio selection method that combines Principal Component Analysis (PCA) with LARS-Lasso regularization, optimized to adapt to the trading constraint of "long-only and no naked short-selling" in Chinas capital market. The core objective of the algorithm is to adaptively select a sparse set of optimal portfolios from high-dimensional asset nodes. By correcting the covariance matrix and the mean return vector via principal components, the portfolio optimization problem is transformed into a linear regression problem with L1 and L2 regularization for solution. Ultimately, hyperparameters are determined through cross-validation, and the empirical performance of the portfolios is evaluated. To clearly describe the algorithm process, the mathematical notations are defined in **Table 3**, along with their corresponding core variables in the code:

Table 3: Notation and Descriptions

Symbol	Meaning	Dimension / Type
T	Number of sample periods (length of time series)	Positive integer
N	Number of asset nodes (feature dimension)	Positive integer
$k_{\min}/k_{\max}$	Minimum / maximum number of non-zero coefficients in the sparse solution	Positive integer
γ	Number of principal components retained by PCA (explaining 95% variance)	Positive integer
λ_0	Mean vector correction coefficient	Positive real number
λ_2	L2 (Ridge) regularization parameter	Positive real number
SR	Sharpe Ratio	Real number
$R \in \mathbb{R}^{T\times N}$	Asset node return matrix	$T \times N$
$\mu \in \mathbb{R}^N$	Mean return vector of asset nodes	$N \times 1$
$\Sigma \in \mathbb{R}^{N\times N}$	Covariance matrix of returns	$N \times N$
$\tilde{\Sigma} \in \mathbb{R}^{N\times N}$	PCA-corrected covariance matrix	$N \times N$
$\tilde{\mu} \in \mathbb{R}^N$	PCA-corrected target mean vector	$N \times 1$
$\beta \in \mathbb{R}^N$	Coefficient vector solved by Lasso (candidate portfolio weights)	$N \times 1$
$w \in \mathbb{R}^N$	Normalized portfolio weight vector	$N \times 1$
$w_{\text{adj}} \in \mathbb{R}^N$	Node depth adjustment weight	$N \times 1$

The essence of the AdaP algorithm is a sparse optimization problem aimed at maximizing the portfolios Sharpe ratio. The original Sharpe ratio is defined as:

$$SR(w) = \frac{w^\top \mu - r_f}{\sqrt{w^\top \Sigma w}} \tag{1}$$

where r_f is the risk-free rate. If the returns are already excess returns, then $r_f = 0$. Because directly maximizing Equation (1) is non-convex and prone to overfitting, the algorithm reformulates the problem using PCA. The steps are as follows:

(1) Eigenvalue Decomposition:

$$\Sigma = VDV^\top \tag{2}$$

where D denotes the diagonal matrix of eigenvalues, and V denotes the matrix of eigenvectors.

(2) PCA Covariance Correction: Retain γ principal components explaining 95% variance:

$$\tilde{\Sigma} = V_\gamma D_\gamma^{1/2} V_\gamma^\top \tag{3}$$

(3) PCA Mean Correction:

$$\tilde{\mu} = V_\gamma D_\gamma^{-1/2} V_\gamma^\top (\mu + \lambda_0 \bar{\mu} \mathbf{1}) \tag{4}$$

where $\bar{\mu} = \frac{1}{N}\sum_{i=1}^{N} \mu_i$ is the average yield rate for all nodes, and $\mathbf{1}$ denotes a vector of all ones.

(4) LARS-Lasso Optimization:

$$\begin{aligned} &\min_{\beta} \frac{1}{2} \left\| \tilde{\Sigma}\beta - \tilde{\mu} \right\|_2^2 + \lambda_2 \left\| \beta \right\|_2^2 + \lambda_1 \left\| \beta \right\|_1 \\ &\text{s.t.} \quad k_{\min} \leqslant \left\| \beta \right\|_0 \leqslant k_{\max} \end{aligned} \tag{5}$$

where λ_1 is automatically generated by the LARS algorithm (as a path parameter), and λ_2 is the Ridge regularization term (implemented via data augmentation). The sparsity constraint requires that the number of nonzero coefficients satisfies $k_{\min} \leqslant \|\beta\|_0 \leqslant k_{\max}$, where $\|\cdot\|_n$ denotes the L_n norm.

To adapt to the "long-only" constraint in the Chinese market, non-negative weight normalization is applied to the solved β:

$$w_i = \begin{cases} \dfrac{\beta_i \cdot w_{\text{adj},i}}{\sum_{j=1}^{N} \max(\beta_j \cdot w_{\text{adj},j}, 0)}, & \beta_i \cdot w_{\text{adj},i} > 0 \\ 0, & \text{otherwise} \end{cases} \tag{6}$$

where $w_{\text{adj},i} = 1/\sqrt[d_i]{\alpha}$ is the node depth adjustment weight (α is the depth weight exponent, set to 2 in this paper; d_i is the depth of node i), ensuring the fairness of weights for nodes at different levels.

3. **Dataset Splitting and Training + Validation Design**

(1) Adopt a basic split of **300 months for training + validation and 60 months for testing:** during the training and validation phase, the test set is completely unseen, and the last 60 data points are selected strictly in chronological order.

(2) After the completion of training + validation, we retrain on the entire training + validation set using the λ_0 and λ_2 obtained from grid search, and then calculate indicators such as annualized return, annualized Sharpe ratio, maximum drawdown, Calmar ratio (annualized return / absolute value of maximum drawdown), and win rate (proportion of periods with positive returns) on the test set. Three main training + validation designs are shown in **Table 4**.

Since the three trainingvalidation splitting methods yield similar performance on the independent test set, we ultimately adopt the **5-Fold Cross-Validation** scheme ($K = 16$, $\lambda_0 = 7.5$, $\lambda_2 = 0.04$), which is the most robust and fully regularized in cross-validation. This choice reduces the models sensitivity to noise and enhances the generalization ability of the strategy.

Table 4: Training + Validation Design

Design	**Description**	**Final Number of Selected Nodes**	λ_0	λ_2
Chronological Split	Strictly in chronological order: the first 240 data points for training, and the last 60 for validation	4	0	0
Time-projection Split	Group every 5 data points in chronological order, with the first 4 assigned to the training set and the remaining 1 to the validation set; this design is more robust to policy effects	5	0.3	0
5-Fold Cross-Validation	Divide the 300 data points evenly into 5 parts in chronological order, use each part as the validation set once, and select parameters based on the average Sharpe ratio	16	7.5	0.04

IV Results

The Adaptive Pruning algorithm successfully reduced the high-dimensional strategy space to a sparse set of 16 effective nodes. For a detailed explanation of the node naming conventions (path interpretations) and the complete list of selected nodes with their corresponding weights, please refer to **Appendix A.1**.

To validate the efficacy of the proposed framework, we compare the APT+AdaP strategy with two benchmarks that represent different levels of information utilization. The first is benchmark 1 (all market/root node), an equal-weighted portfolio of the entire stock universe. This serves as a proxy for the broad market performance and tests the fundamental validity of the tree-based segmentation. The second is benchmark 2, which applies an equal-weighting scheme to the 16 selected nodes. By comparing the APT+AdaP strategy against this benchmark, we aim to isolate the marginal contribution of the Lasso-based weighting mechanism from the node selection process, thereby verifying that the performance is not merely driven by selecting high-quality assets but also by the robust allocation of weights.

Figure 1 below illustrates: (a) the cumulative return comparison between the APT strategy and the benchmark portfolios; (b) the monthly return distribution of the APT strategy; (c) the portfolio drawdown of the APT strategy; (d) the top ten node weights in the final APT strategy portfolio.

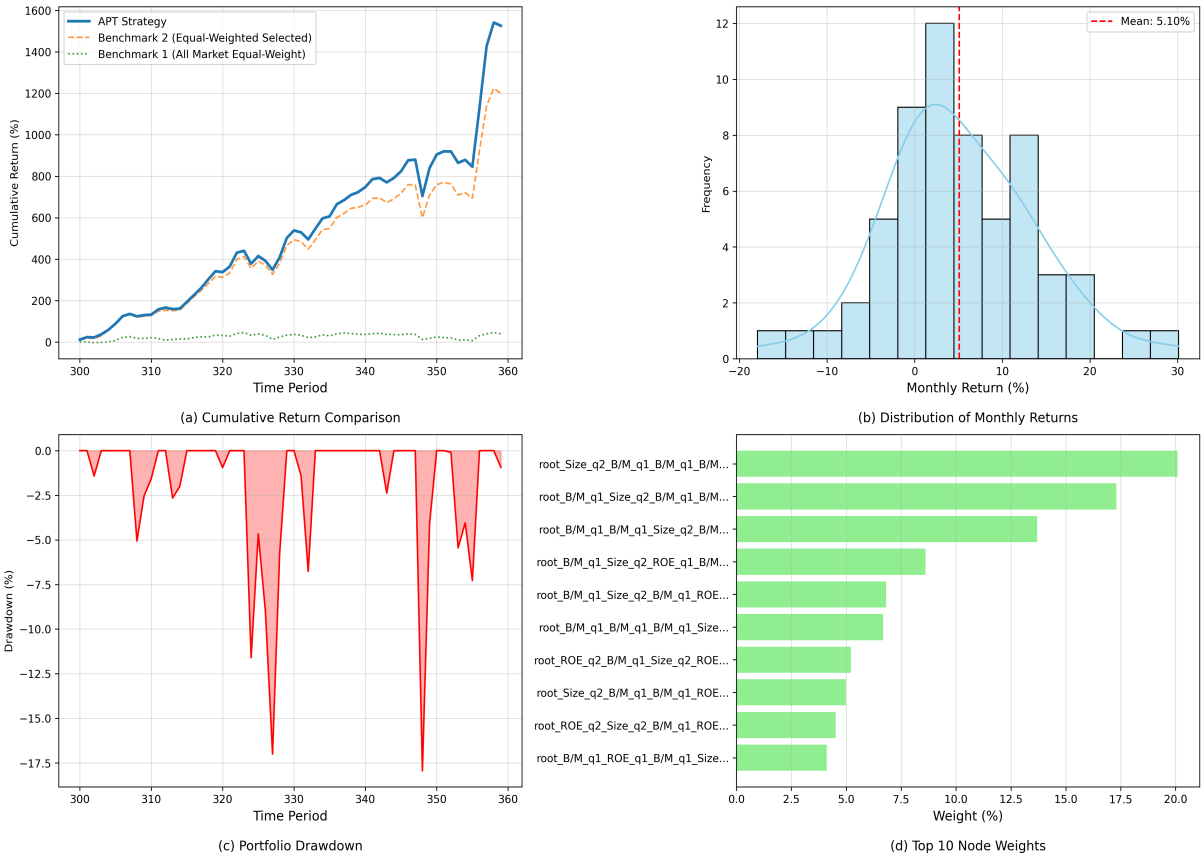


Figure 1: Comparison of cumulative returns

Table 5: Out-of-Sample Performance

Indicator	**APT+AdaP**	**Benchmark 1**	**Benchmark 2**
Annualized Return	81.64%	9.54%	73.63%
Annualized Volatility	29.66%	22.37%	29.43%
Sharpe Ratio	2.75	0.426	2.50
Cumulative Return	1526.35%	39.61%	1200.24%
Maximum Drawdown	−17.94%	−27.03%	−18.56%
Calmar Ratio	4.55	0.353	3.97
Monthly Win Rate	75.00%	55%	73.33%

As shown in **Table 5**, the strategy delivers exceptional profitability, achieving a cumulative return of over 1500% and surpassing both benchmark 2 (equal weight on selected nodes) and benchmark 1 (equal weight on all market). This highlights two key value sources: effective node screening and the pruning algorithm's adaptive weighting, which further enhances performance beyond simple equal weighting.

The strategy also shows strong risk return characteristics. Monthly returns exhibit the maximum drawdown is limited to −17.94%. Combined with the high return profile, this yields a Calmar ratio of 4.55, indicating excellent risk adjusted performance.

Finally, the resulting portfolio is highly sparse with clear economic interpretation. The pruning algorithm concentrates weights on a small set of core nodes, with the top weighted exposures reflecting interactions between size and value factors, specifically deep value stocks within mid to large cap firms. This pattern provides both statistical robustness and coherent economic rationale for the strategy's excess returns.

V Conclusion

This paper introduces a two-stage portfolio construction framework integrating adaptive portfolio trees and an adaptive pruning algorithm, designed to address the limitations of traditional linear factor models in characterizing nonlinear relationships and high-dimensional interaction structures. By leveraging APT to generate a candidate portfolio space via multi-layer feature splitting and applying AdaP to isolate sparse, stable representative nodes, the methodology successfully balances model interpretability with predictive power. Empirical results from out-of-sample testing demonstrate the framework's superior efficacy, achieving an annualized Sharpe ratio of 2.75 and a cumulative return exceeding 1500%, while limiting the maximum drawdown to only -17.94%. These metrics indicate that the proposed strategy significantly outperforms benchmarks in both capital appreciation and risk control capabilities.

The structural characteristics of the pruned node set reveal a consistent and economically meaningful pricing mechanism within the Chinese equity market. Specifically, the book-to-market ratio (B/M) acts as the dominant splitting criterion, with the highest-weight portfolios consistently originating from the low B/M quartile, underscoring the strong explanatory power of the value factor. Market capitalization serves as the secondary layer, forming a core "value-size" path where large-cap value stocks contribute the most significant and stable risk premiums. Furthermore, return on equity (ROE) emerges in deeper splitting layers to distinguish profitability characteristics within the core clusters. The resulting weight distribution is heavily concentrated on this primary path, while lower-weight peripheral nodes provide essential supplementary explanations for tail risks and extreme return divergences, thereby enhancing the model's overall robustness.

Future research may expand upon these findings by incorporating a broader set of features, such as momentum and high-frequency volatility, to explore diverse factor combinations. Additionally, applying this framework to alternative asset classes and international markets will serve to validate its generalizability. There is also significant potential in hybridizing tree structures with deep learning architectures, such as tree-enhanced transformers or graph neural networks, to capture complex dynamic interactions and automate regime-dependent weight adjustments. In summary, the APT+AdaP framework offers a rigorous, interpretable, and high-performance approach to nonlinear asset pricing, establishing a promising foundation for the design of quantitative trading strategies.

Appendix

A.1 Sparse and Interpretable Portfolio

A.1.1 Node Naming Explanation

Each retained node is represented by a path string in the format "`root_<Feature1>_qX_<Feature2>_qY_...`"„ indicating the child node obtained by sequentially splitting `Feature1` at Level 1, `Feature2` at Level 2, etc., starting from the root node.

- `q1`: The normalized value of the feature falls **below the 0.5th quantile (smaller)**;
- `q2`: The normalized value of the feature falls **above the 0.5th quantile (larger)**.

All features are standardized/normalized and then binned by the sample median (0.5th quantile), with node names concatenated in the order of splitting.

A.1.2 Selected Nodes and Their Weights

The following table lists the nodes retained after pruning and their corresponding weights, sorted by the original weight order.

Table A1: List of Selected Nodes and Weights

Node Path	**Weight**
root_Size_q2_B/M_q1_B/M_q1	0.200851892600014
root_B/M_q1_Size_q2_B/M_q1_B/M_q1	0.173002464964539
root_B/M_q1_B/M_q1_Size_q2_B/M_q1	0.136989919268715
root_B/M_q1_Size_q2_ROE_q1_B/M_q1	0.086136344071943
root_B/M_q1_Size_q2_B/M_q1_ROE_q1	0.068157913420469
root_B/M_q1_B/M_q1_B/M_q1_Size_q2	0.066754927437884
root_ROE_q2_B/M_q1_Size_q2_ROE_q2	0.052072636029621
root_Size_q2_B/M_q1_B/M_q1_ROE_q1	0.049769145886026
root_ROE_q2_Size_q2_B/M_q1_ROE_q2	0.045209218529220
root_B/M_q1_ROE_q1_B/M_q1_Size_q2	0.041101144601902
root_B/M_q1_B/M_q1_Size_q2_ROE_q1	0.029256923640976
root_Size_q2_B/M_q1_ROE_q1_B/M_q1	0.016500614235560
root_Size_q2_ROE_q1_B/M_q1_B/M_q1	0.013833413634589
root_Size_q2_B/M_q1_ROE_q1_ROE_q1	0.007846366247768
root_B/M_q1_Size_q2_ROE_q2_ROE_q2	0.006428015435482
root_B/M_q1_ROE_q2_Size_q2_ROE_q2	0.006089059995292
Sum	**1.000000000000000**

Note: The sum of the above weights equals 1.

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